Session Layer & Its Functions

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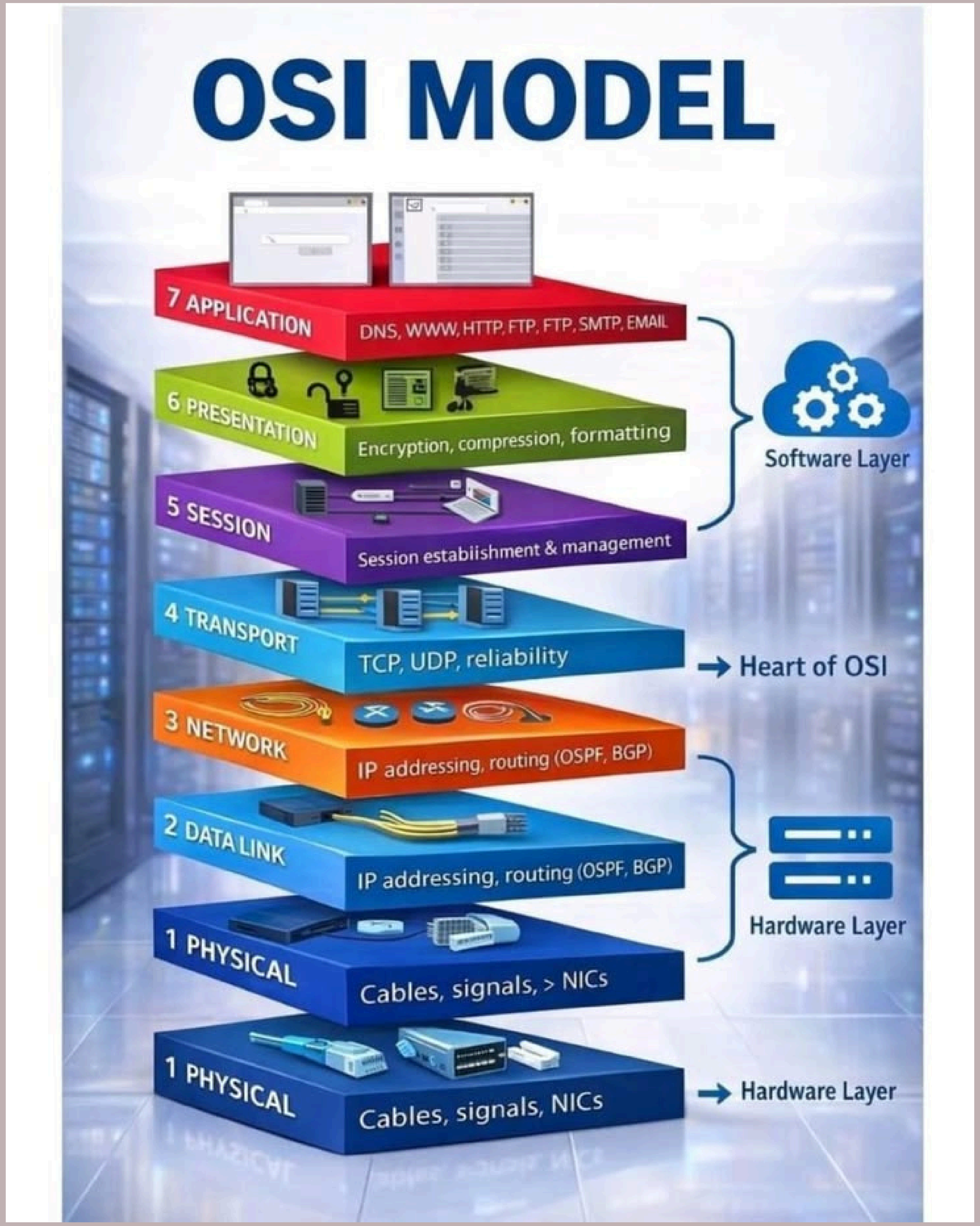
Overview of the OSI Model



The OSI model comprises **seven distinct layers**, each serving a unique purpose in data communication. Layer 5, the Session Layer, manages connections and controls sessions between applications, ensuring smooth data exchange.

SESSION LAYER

Introduction to Session Layer (OSI Layer 5)

- The Session Layer is the 5th layer of the OSI model.
- It manages communication sessions between two computers or applications.
- Responsible for starting, maintaining, and ending connections.
- Ensures smooth data exchange during communication.
- Provides synchronization and recovery if connection is interrupted.

Core Functions of the Session Layer

01 **Session Establishment**

Initiates and manages communication between systems.

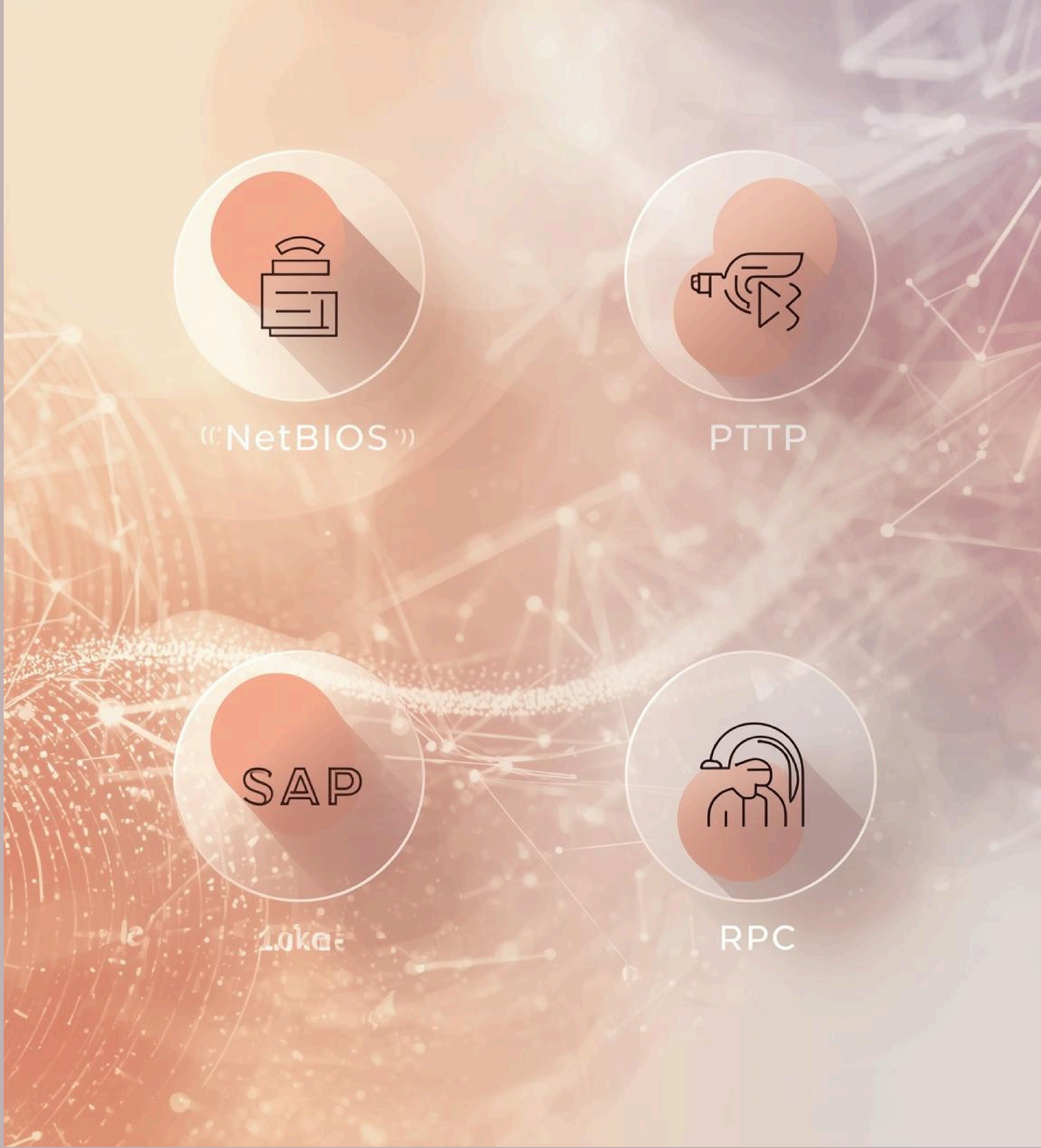
02 **Dialog Control**

Regulates data exchange direction and flow.

03 **Synchronization**

Implements checkpoints for session recovery processes.

Key Protocols of the Session Layer



The Session Layer utilizes various protocols, such as **NetBIOS for network communication**, PPTP for secure tunneling, SAP for session announcements, and RPC for remote procedure calls, ensuring effective data exchange.

Dialog Control

Understanding Communication Modes

Half-Duplex

Half-duplex communication allows data transmission in **one direction at a time**, meaning that devices take turns to send and receive messages, ensuring orderly communication but limiting efficiency.

Full-Duplex

Full-duplex communication permits simultaneous data transmission in **both directions**, enabling devices to send and receive messages concurrently, which enhances efficiency and responsiveness in interactions.

Synchronization and Recovery in Sessions

The session layer employs **checkpoints** to restore connections seamlessly after interruptions, ensuring that data is preserved and preventing **loss or duplication**, which is crucial for maintaining communication integrity.

Real-World Applications of the Session Layer

- **Video Calls (Zoom, Google Meet): Starts, manages and ends call sessions; reconnects if network drops.**
- **Online Banking: Maintains secure login session and logs out after inactivity.**
- **Web Browsing (Gmail, Instagram): Keeps user logged in until logout.**

The Importance of the Session Layer

The session layer **enables reliable communication** across networks, facilitating organized multi-session management while forming a critical foundation for higher-level application protocols and ensuring efficient data exchange.

Thank You